

CONFIDENT-TAIL TEST-TIME ADAPTATION MOVES THE THRESHOLD, NOT THE RANKING, IN AUDIO DEEPPAKE DETECTION

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ABSTRACT

Cross-corpus failure in audio deepfake detection is often a failure of the decision threshold rather than of score ordering: a released detector in our In-the-Wild evaluation has ROC-AUC 0.9171 but 39.08% accuracy at threshold 0.5. Confident-tail self-training, which pseudo-labels the lowest- and highest-scoring target clips, exploits the ordering that survives. Over 79 adaptation runs spanning four released checkpoints, two corpora, spoof prevalence 0.01 to 0.972 and budgets 0.02 to 0.50, it proves to be almost purely an operating-point procedure: the accuracy its scores can attain shifts by a median of 0.12 points while the gap between that ceiling and the accuracy obtained at threshold 0.5 shifts by 14.87, and in no run does the ceiling move further. Where the new operating point lands follows a counting quantity. A bucket assigned class c with realized fraction b_c of a pool with prior π_c cannot be purer than $\min(1, \pi_c/b_c)$, so a fraction $\Delta = \sum_c (b_c - \pi_c)^+$ of the pool is necessarily mislabeled. The post-adaptation gap increases with Δ in 12 of the 14 strata that vary the budget, and adaptation improves accuracy exactly when that gap ends below the source’s, on 79 of 79 runs. Purity orders none of this, and neither does the budget: $q = 0.50$ helps at 37% spoof prevalence while $q = 0.30$ destroys accuracy at 97%, and on a balanced pool a perfectly pure selection still costs 9.6 ± 0.8 points over three seeds.

Index Terms— audio deepfake detection, test-time adaptation, label shift, pseudo-labels

1. INTRODUCTION

Self-supervised speech representations paired with anti-spoofing back-ends provide strong audio deepfake detectors [1, 2], but transfer across recording conditions and synthesis systems remains difficult [3]. A target-domain failure can lie in the score ordering, in the decision threshold, or in both: a released detector in our In-the-Wild evaluation has ROC-AUC 0.9171 but 39.08% accuracy at threshold 0.5.

Test-time adaptation (TTA) updates a source model from unlabeled target observations, and both entropy minimization [4] and pseudo-label self-training can reinforce incorrect

predictions. We examine one self-training rule, assigning the bottom and top score quantiles to bona fide and spoof, and ask what it actually changes. It is not a rule proposed elsewhere: it is our own prior design, an instance of confidence-based pseudo-label selection [5, 16] with a fixed budget per tail. We find it changes the operating point and little else, and that a counting constraint on the budget predicts where that point lands.

Class imbalance in adaptation is established ground: Label Shift Adapter [6] adapts parameters to estimated target proportions, RLSbench [7] studies joint label and conditional shift, black-box shift estimation (BBSE) [8] recovers target priors from a source confusion matrix, and T²A [9] treats deepfake TTA by uncertainty-aware negative learning. Our claim is narrower and empirical: what this rule changes, and the counting quantity that predicts it.

2. TAIL SELECTION AND ADAPTATION

Let $y = 1$ denote spoof, $s_\theta(x) \in [0, 1]$ the detector’s spoof score, and the unlabeled adaptation pool have N clips with unknown class fractions π_c . Target labels are used only for evaluation, controlled resampling and explicitly marked oracle analyses.

2.1. A necessary condition for pure pseudo-labels

A bucket C_c assigned pseudo-label c , with realized fraction $b_c = |C_c|/N > 0$, contains at most $\pi_c N$ correct examples. Thus

$$\text{purity}(C_c) \leq \min\{1, \pi_c/b_c\}. \quad (1)$$

This is a finite-pool counting bound and makes no assumption about the model. An update, however, does not consume a purity: it consumes wrongly labeled examples. The same counting fact says that the selection necessarily mislabels a fraction

$$\Delta = \sum_c (b_c - \pi_c)^+ \quad (2)$$

of the pool. Purity is a within-bucket rate and Δ a share of the pool, and the two order configurations differently: a small

bucket can have very low purity while mislabeling almost nothing. Section 4 finds that Δ orders the observed outcomes and that purity does not.

For exact symmetric tails, $b_0 = b_1 = q$, both statements reduce to $\Delta = (q - \min\{\pi, 1 - \pi\})^+$, so perfect purity is possible only if

$$q \leq \min\{\pi, 1 - \pi\}, \quad \pi = \pi_1. \quad (3)$$

At 97.22% spoof prevalence a bucket labeling 30% of the pool bona fide has purity at most 9.27%, and no amount of AUC removes that. Feasibility is not sufficiency: score overlap contaminates feasible tails, and learning tolerates impure labels when there are few of them. That is why we test (2) rather than (3).

A prior allocation uses nominal fractions $q_c = 2q\hat{\pi}_c$, keeping the total selection fraction $2q$. With exact buckets and a correct prior, $q \leq 1/2$ ensures $q_c \leq \pi_c$; with estimated priors it instead requires $\hat{\pi}_c/\pi_c \leq 1/(2q)$, a limit on overestimation rather than a safety theorem. A smaller symmetric q needs no point estimate, but a guaranteed feasible nonzero value still needs a lower bound on the minority prevalence.

Quantile ties. Our runs use inclusive thresholds, $s \leq Q(q_0)$ and $s \geq Q(1 - q_1)$, with spoof assignments taking precedence on overlap, so realized sizes differ from nominal budgets. Equations (1) and (2) are evaluated at the realized b_c throughout; we never read a nominal budget as an exact sample count.

2.2. Prior estimation and objective

For $M_{ij} = P_S(\hat{y} = j \mid y = i)$ and a target prediction histogram h from source predictions on the adaptation pool, BBSE solves $M^\top \hat{p} = h$. We estimate M from up to 4,000 class-balanced clips of the checkpoint’s training corpus, clip to $[10^{-4}, 1 - 10^{-4}]$ and renormalize; using training rather than held-out data makes its error rates optimistic.

At each of four passes, the current model scores the adaptation pool and refreshes pseudo-labels. For a mini-batch B and its selected subset B_C ,

$$\begin{aligned} \mathcal{L}_B = & \frac{1}{|B_C|} \sum_{x \in B_C} \text{CE}(p_\theta(x), \tilde{y}_x) \\ & + \frac{\lambda}{2|B|} \sum_{x \in B} \|p_\theta(a(x)) - \text{sg}[p_\theta(x)]\|_2^2, \end{aligned} \quad (4)$$

where $p_\theta = (1 - s_\theta, s_\theta)$, the CE term is omitted when B_C is empty, and $\lambda = 0.3$. The consistency term uses all batch examples and both probability components. The perturbation is per-clip gain sampled uniformly from $[0.7, 1.3]$ plus Gaussian noise with standard deviation 0.005. Only the last four transformer blocks’ LayerNorm affine parameters and the classifier head are updated; BatchNorm statistics stay frozen. Adam uses learning rate 10^{-4} and batch size 32. The released DeepFense models have 16,706 trainable parameters under

Table 1. The 400,435 available DF eval trials, DF-2, adaptation seed 0, whose always-spoof accuracy baseline is 97.22%. Threshold 0.5 except in the two frozen-source controls. All columns but AUC are percentages; G_τ is an oracle diagnostic and smaller is better.

Method	EER	AUC	Acc.	BA	G_τ
Source	4.51	0.9923	98.87	88.28	0.20
Entropy (Tent-style)	4.28	0.9765	98.52	73.51	0.57
Fixed confidence 0.95	4.30	0.9918	99.02	85.03	0.00
Symmetric $q = 0.02$	4.19	0.9930	98.96	83.88	0.01
Symmetric $q = 0.3$	5.84	0.9846	62.96	80.89	35.38
BBSE tail budget	4.06	0.9931	98.95	82.34	0.02
Median threshold only	4.51	0.9923	59.06	78.94	40.01
BBSE threshold only	4.51	0.9923	98.88	88.19	0.19

this rule. A fixed-confidence control instead selects scores at most 0.05 or at least 0.95, with the same objective.

3. EVALUATION PROTOCOL

Checkpoints and pools. We use DeepFense XLS-R 300M + AASIST checkpoints trained on ASVspoof2019-LA with seeds 2, 42 and 240 [10, 11], denoted DF-2, DF-42, DF-240, and a WaveFake-trained SSL-AASIST checkpoint WF [12]. Audio is 16 kHz, cropped to 64,000 samples for DeepFense and 64,600 for WF.

The official ASVspoof2021-DF metadata has 533,928 eval trials among 611,829 across three phases [13]. Our available *eval-only* subset holds 400,435 of them, of which 389,299 are spoof and 11,136 bona fide, so 97.219% spoof, covering 75.00% of the spoof class and 74.89% of bona fide. The missing shard rules out a full-set comparison; all comparisons use the same trials.

The main DF run adapts on 20,000 clips with adaptation seed 0; we report the available pool and its 380,435-clip disjoint remainder. Seeds 1 and 2 adapt 20,000 clips of a fixed 50,000-trial subset (seed 12345), sensitivity checks on overlapping data rather than nine independent datasets. In-the-Wild uses all 31,779 clips (11,816 spoof, 37.182%) with 20,000 adaptation clips.

Prevalence intervention. We resample either corpus to a target spoof proportion (seed 4242) by thinning whichever class must shrink: In-the-Wild to eight priors from 0.01 to 0.97 including its native 0.372, and DF to 0.01, 0.10, 0.50 and 0.90, the last a balanced pool. Every resampled pool adapts on 8,000 clips, so the optimization budget is identical at every prevalence and a disjoint remainder always remains; the evaluation pool size still varies with prevalence, which the experiment does not separate from it.

Metrics and controls. We recompute metrics from saved float32 probabilities. ROC-AUC assigns half credit to ties; EER linearly interpolates the crossing of FPR and FNR on the ROC, grouping identical scores. Interpolated EER need

not correspond to an attainable deterministic threshold. We also report accuracy, balanced accuracy, and the oracle threshold gap $G_\tau = \max_t \text{Acc}(s, t) - \text{Acc}(s, \tau)$, whose maximum ranges over all distinct thresholds including constant predictions. It uses evaluation labels and is a diagnostic of threshold headroom, not a deployable method or a calibration error.

Two label-free controls keep source scores fixed and pick a threshold from adaptation scores: their median, or their $(1 - \hat{\pi})$ -quantile using BBSE. Both preserve AUC and EER; neither is a claim of Bayes optimality.

We additionally report fixed-configuration entropy adaptation (Tent-style) under the same update settings [4].¹

4. RESULTS

4.1. Adaptation moves the operating point, not the ranking

Across all 79 symmetric-budget runs – four checkpoints, two corpora, spoof prevalence 0.01 to 0.972, budgets q from 0.02 to 0.50 and three adaptation seeds on eight cells – adaptation leaves almost untouched the accuracy its scores can attain, and moves only where the decision sits. The best accuracy attainable over all thresholds shifts by a median of 0.12 points; the gap between that ceiling and the accuracy obtained at threshold 0.5 shifts by 14.87 points, and in no run does the ceiling move further than the gap (Fig. 1(a)). ROC-AUC shifts by a median of 0.12 points. This is why the metrics disagree so violently: the same DF-2 run that changes accuracy by 35.9 points changes EER by 1.33.

4.2. Contamination sets where the operating point lands

The quantity that predicts the new operating point is Δ . In 12 of the 14 checkpoint/prevalence strata that vary the budget, the post-adaptation threshold gap increases with Δ (Fig. 1(b)), and in the seven whose source threshold is already well placed – a source gap under 17 points – it does so strictly, six of them at $\rho = +1$. The two exceptions both have badly placed sources, DF-42 at $\pi = 0.01$ leaving 95.1 points on the table: there the smallest budgets cannot move the model at all and larger ones close the gap rather than open it.

¹Its scores saturate: 98.07% lie within 10^{-6} of 1.0 but only 93.02% exactly at 1.0, and conflating the two makes its AUC look impossible. With exact ties grouped it reaches 4.28% EER and 0.9765 AUC against the source’s 4.51% and 0.9923, a gain near one operating point coexisting with worse integrated ranking. The scoring convention matters too: ROC interpolation gives the ETA ablation 12.31% EER where a nearest-point rule gives 7.44%. Three further published TTA methods were run the same way, as controlled variants rather than tuned reproductions: an ETA reliability-only ablation without redundancy or Fisher terms (EER 12.31%, and an all-spoof decision at threshold 0.5 on DF-42 and DF-240), SAR with recovery threshold scaled to $(0.2/\ln 1000)\ln 2$ (4.48%), and IM-PL, a SHOT-inspired information-maximization control with probability-space centroid pseudo-labels that freezes the head and updates 16,384 parameters (6.35%) [14–16]. None is a tuned reproduction and none establishes anything about those methods’ published performance; their audited rows are in the released tables.

With the invariance above, this says exactly when the rule is worth applying: adaptation raises accuracy if and only if its post-adaptation gap ends below the source’s, which holds on 79 of 79 runs. Neither purity nor the budget orders the same outcomes. At the native In-the-Wild prior $q = 0.50$ – the largest symmetric budget there is, labeling the entire pool – leaves a 7.43-point gap and gains 7.47 points of balanced accuracy, while $q = 0.15$ at $\pi = 0.01$ leaves 28.17 and gains 2.64; a tail of purity 0.107 gains 9.36 points where one of purity 0.320 loses 0.37. Nor is a clean selection sufficient: on a balanced DF pool, where $\Delta = 0$ and the measured purity is 1.000, adaptation still moves the gap from 7.38 to 17.4 ± 0.9 and costs 9.6 ± 0.8 points of accuracy across three seeds.

4.3. Anchoring on ASVspoof2021-DF

On DF-2 (Table 1) symmetric $q = 0.3$ raises EER from 4.51% to 5.84%, lowers balanced accuracy from 88.28% to 80.89% and accuracy from 98.87% to 62.96%, against a 97.22% always-spoof baseline and a 35.38-point threshold gap. Allocating the tails by a BBSE prior estimate instead reaches 4.06% EER and 98.95% accuracy, but 82.34% balanced accuracy, below the source: one effective remedy, not a uniformly best one, and not an improvement for both classes.

The same pattern holds on DF-42 and DF-240. Across the nine checkpoint/run combinations BBSE has lower EER than the symmetric budget in 9/9, and than the source in 8/9 on the available pool and 7/9 under disjoint evaluation. A paired stratified bootstrap over trials (2,000 replicates, resampled within class so the prior stays fixed) separates the two: all nine symmetric differences have 95% intervals excluding zero, but only 4/9 and 5/9 of the source differences do, and DF-240’s is -0.02 points $[-0.14, +0.12]$. These counts are descriptive, not independent evidence.

Re-thresholding alone reproduces that accuracy: a frozen-source BBSE threshold gives 98.88% against the source’s 98.87%, while the median gives 59.06%.

4.4. Corpus shift and prior estimation

At the native In-the-Wild prior the budget has no counting obstruction and improves accuracy on three of four checkpoints. For DF-42 it raises AUC from 0.9171 to 0.9473 and accuracy from 39.08% to 82.39%, though a frozen-source median threshold already reaches 78.10%. BBSE estimates 97.9% spoof on this 37.2%-spoof corpus and lands at 14.54% EER and 47.99% accuracy, against 11.59%/91.28% for a non-deployable oracle. BBSE is identifiable under label shift with an invertible confusion matrix [8], but under unrestricted conditional shift $P_{\mathcal{T}}(\hat{y} = 1) = \pi \text{TPR}_{\mathcal{T}} + (1 - \pi) \text{FPR}_{\mathcal{T}}$ cannot determine all three unknowns, which is why a source confusion matrix may fail to transfer.

Table 2. WF on In-the-Wild resampled to prevalence π , EER%/balanced accuracy%. BA has the same 50% trivial baseline in every row, where the always-spoof accuracy baseline moves from 90.00% to 99.00%. A dash marks a budget not run at that prevalence.

π	N	Source	$q = .02$	$q = .10$	$q = .30$	BBSE
0.372	31,779	5.06/82.7	—	—	4.55/95.4	4.54/95.5
0.010	20,165	5.10/82.7	4.95/92.4	5.10/92.1	5.27/75.7	5.17/82.2
0.020	20,370	5.16/82.7	5.00/92.9	5.12/91.8	5.23/79.0	5.16/82.7
0.050	21,014	5.14/82.6	5.04/93.7	5.11/94.6	5.09/78.8	5.10/82.6
0.100	22,181	5.12/82.7	5.04/94.0	4.96/94.5	5.05/82.3	5.02/85.2
0.372	31,778	5.06/82.6	—	—	4.87/95.2	—
0.900	13,129	5.26/82.7	5.04/93.7	5.26/95.0	6.02/84.4	5.03/93.7
0.950	12,438	4.66/83.6	4.58/92.5	4.94/94.6	5.66/83.0	4.52/91.6
0.970	12,181	3.74/83.5	3.91/93.3	4.66/93.7	5.21/81.8	3.84/92.3

4.5. Intervening on prevalence

On resampled In-the-Wild (Table 2) the same holds. At 97% spoof $q = 0.3$ gives accuracy 65.42% against the source’s 98.94%, but balanced accuracy 81.8% against 83.5% and EER 5.21% against 3.74%; $q = 0.02$ gains 9.8 points of balanced accuracy for 0.17 of EER.

The deployment direction. Real pools are mostly bona fide, so we extend the sweep to $\pi \in \{0.01, 0.02, 0.05, 0.10\}$, where the spoof tail is the constrained one. The mirror image appears: at $\pi = 0.01$, $q = 0.3$ takes accuracy from 65.72% to 51.90% and balanced accuracy from 82.69% to 75.71%, with the damage carried entirely by false alarms (FPR 48.58% against 34.62%, FNR 0.00% in both). A small budget does not break where (3) says it must: its spoof tail is bounded at 52.7% purity and measured at 33.3%, yet reaches 97.00% accuracy. Every arm at every prevalence has EER between 4.95% and 5.27% while accuracy spans 51.90% to 97.00%.

Budget sweeps. Sweeping q at fixed prevalence traces the relation directly, and repeating three points of each sweep over three adaptation seeds shows the ordering is not seed noise. At $\pi = 0.01$ balanced accuracy falls from $+12.00 \pm 0.17$ at $\Delta = 0.049$ through $+3.35 \pm 0.62$ to -5.19 ± 1.63 at 0.289; at $\pi = 0.97$ it falls from $+11.62 \pm 0.10$ through $+5.43 \pm 0.46$ to -1.80 ± 0.20 . Each step between budgets is five to sixteen times the seed spread, and both orderings survive disjoint evaluation.

Prior estimation in that direction. BBSE, which repaired the high-prevalence collapse, does not transfer: it estimates 35.9–41.8% spoof on pools that are 1.1–9.7% spoof and changes accuracy by -1.0 to $+4.7$ points, where a prior-free $q = 0.02$ gains 27.1 to 31.3.

5. DISCUSSION AND LIMITATIONS

On these detectors, confident-tail adaptation chooses an operating point rather than improving a detector, which is a narrower reading than the rule is usually given and an easier one

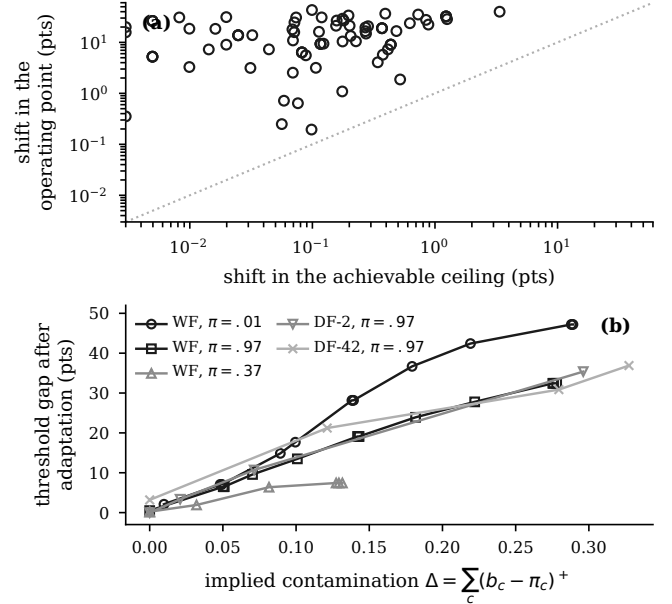


Fig. 1. Over 79 runs spanning four checkpoints, two corpora, prevalence 0.01–0.972 and budgets 0.02–0.50. (a) How far adaptation moves the attainable accuracy ceiling against how far it moves the operating point; every run lies above the diagonal, and values below 3×10^{-3} points are clipped to the axis. (b) The post-adaptation threshold gap against the contamination (2) implies.

to deploy responsibly: use it where a source threshold is believed to be badly placed, and bound Δ otherwise. The counting obstruction is what makes Δ computable, and it is useful only in that currency: a purity ceiling orders nothing, while the share of the pool that must be mislabeled orders the resulting operating point wherever the source threshold was worth preserving. It bounds the damage and does not promise a benefit; a balanced pool with a perfectly pure selection still lost 8.6 points.

The evidence is four released checkpoints from one architecture family, two corpora, an incomplete DF eval partition, one adaptation recipe, and three adaptation seeds on eight of the cells, on pools that share audio within a corpus. Δ uses realized bucket sizes and the true prior, so it is a diagnostic, not a label-free rule; making it one needs a prior estimate, and ours fail exactly where the deployment cases lie.

Reproducibility and assistance. `audit_icassp.py` verifies every saved score array against its historical AUC and accuracy, recomputes every reported metric and the selection audit, and writes the tables and input hashes, with no neural training or inference. Code, scores and outputs: https://github.com/RohamIzaidooost/Kaggle_deepFakeAudio. The study uses existing public data and checkpoints, with no new human-subject data. AI coding assistants contributed to the text, code, figure and experiment scripts under the authors’ direction.

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